

What makes us Tick?

That's the question scientists are trying to answer while working on the human gene, aided by supercomputers and biotechniques

The world is filled with nearly 6 billion people, each different from the other and yet sharing a lot of traits in common. These traits are handed down from one generation to the next through genes, the units of information inside the cells of your body.

What if we can modify the gene itself for a stronger build or affect the gene such that it is more resistant to diseases? How does a set of instructions work to create humans and other forms of life? A quest to the answer of these questions led to the formation of the Human Genome Project (HGP) in 1990. This project is coordinated by the United States Department of Energy and National Institutes of Health. It was initially scheduled to last for a period of 15 years.

More than 18 countries are now participating in this quest which involves identifying the genes that are believed to make up the human DNA. The

Human Genome Organization (HUGO) helps to coordinate the widespread international collaboration in the genome project.

Where does it stand today?

Numerous sections of the genome map have been completed and an entire human genome sequence, together with its analysis was released in February this year. The outcome from HGP would comprise a resource of detailed information about the structure, organisation and the functionalities of the human DNA. But let's first understand what a genome is.

The nucleus of each cell in the human body contains the set of genes that provide a complete set of instructions for making you you—sex, colour of eyes, physical structure and even your temperament and talents. This set of codes is the genome. The information of the genes is organised into strings of DNA called chromosomes. Each chromosome carries several thousand genes. Identifying on which chromosome a specific gene is located and where along that chromosome it can be found is called 'mapping'.

DNA is made up of four similar chemicals (called bases or nucleotides), which are abbreviated as A, T, C, and G. These bases are repeated millions of times, as combinations, throughout a genome. The human genome for instance has 3 billion base pairs that are arranged in an order that's unique to each individual. The process of finding out the exact order in which the base pairs occur is called 'DNA sequencing'.

As you can imagine, it is a Herculean task. It is estimated that a hundred thousand patterns of the DNA would be revealed by simulating the process of DNA sequencing using the information already gathered in the databases.

Computers and biotechnology

Considering the sheer volume of data that researchers working on the HGP have to deal with, computers and other advanced electronic technology are an essential part of the Project.

Laboratory automation involves the use of robots to handle the genetic material along a 'production line'. Electro-mechanical systems, lasers, sensors, and sophisticated computers all play key roles in analysing and recording the results. For speed, efficiency and accuracy, computers are used in the process of DNA sequencing, which involves reassembling small fragments of DNA into a complete sequence to find the shortest sequence containing all the fragments.

Dynamic programming is used during the

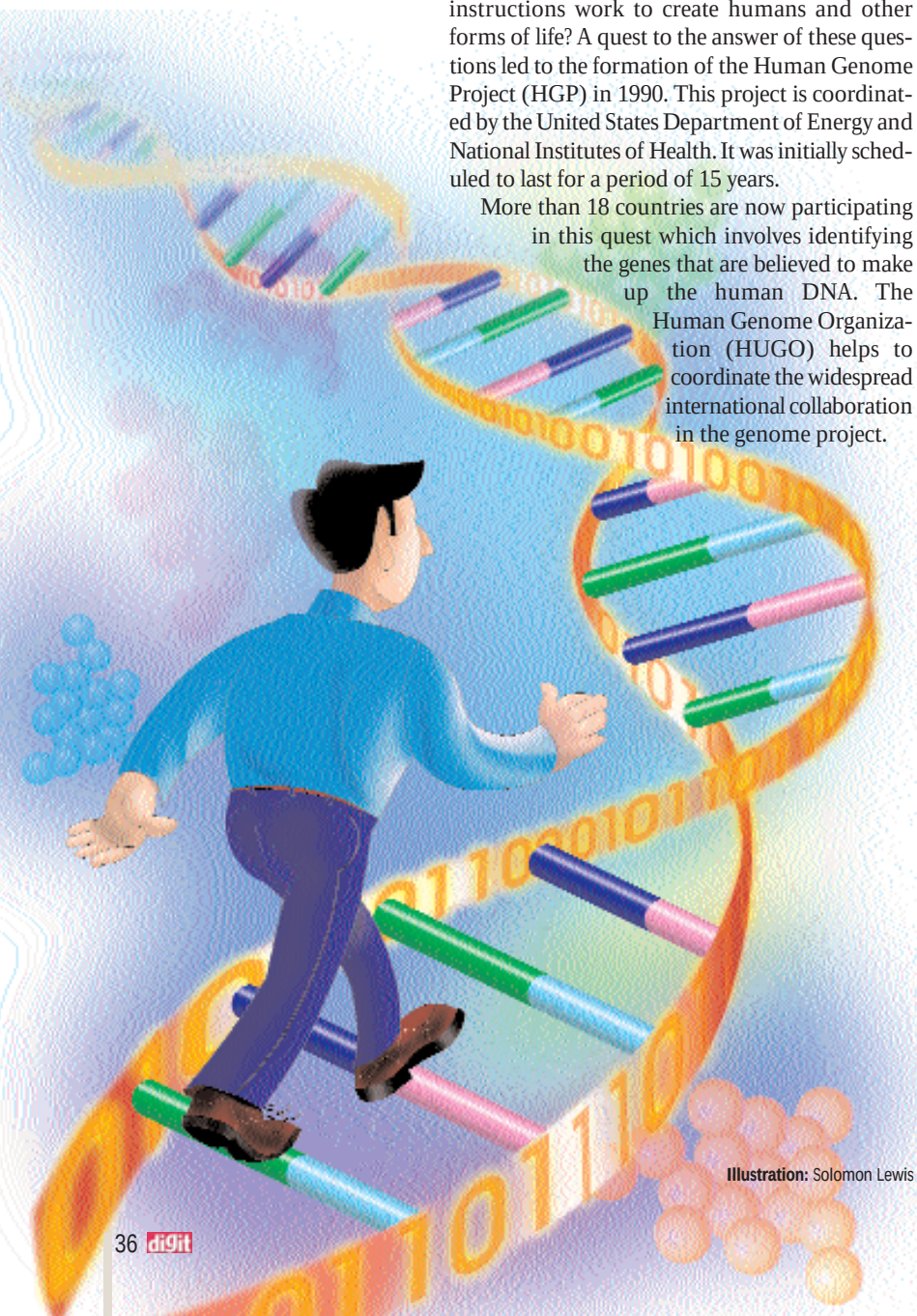


Illustration: Solomon Lewis